

View results

Respondent

55

Anonymous

312:35

Time to complete

A. General information about the respondent

1. * Country

Timor-Leste



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

National Institute of Science and Technology (INCT)

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

www.inct.gov.tl

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The National Institute of Science and Technology (INCT) is a public institute, endowed with legal personality, broad administrative, financial and patrimonial autonomy, as well as scientific and editorial autonomy, without prejudice to the State's supervisory action. It was created in 2014 by Decree-Law no. 23/2014, of September 3, which was amended for the first time by Decree-Law no. 5/2023, of March 8.

As a public entity, the INCT has the following missions stipulated in article 4 of the decree-law:

☐ To continuously promote the advancement of scientific and technological knowledge in Timor-Leste, exploiting the opportunities that reveal themselves, in all scientific and technological fields, with the potential to reach the highest international standards of knowledge creation and to stimulate its dissemination and practical application as a factor in the development and improvement of the well-being of the population.

☐ Actively seeks the continuous promotion of science, innovation and technology in East Timor, as well as storing, preserving and disseminating the intellectual heritage of a scientific, technological and academic nature and stimulating and promoting scientific research in the country.

7. * Full name of officially designated contact person(s) who completed this survey

Benvindo Domingos José Guterres

8. * Position of officially designated contact person(s) who completed this survey

National director of Cataloging and Repository Direction

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

José Cornélio Guterres, President of the INCT Executive Board

11. Brief description of the consultation process established for the preparation of the report

The National Institute of Science and Technology (INCT), in collaboration with UNESCO Jakarta and the National Commission of UNESCO Timor-Leste, organized a two days' workshop on Open Science on Feb 17-18,2025 in Dili, Timor-Leste. The event was officially opened by the Minister of Higher Education, who emphasized the importance of Open Science in Timor-Leste, highlighting its role in promoting knowledge sharing, innovation, and collaboration among researchers. Following the opening, various stakeholders presented on the state of Open Science in the country, addressing existing challenges. The workshop also included a discussion session on Open Change in Timor-Leste, where participants exchanged ideas on how to advance Open Science and improve accessibility to scientific knowledge. The event concluded with an explanation on how to fill out the survey questionnaire, ensuring that participants understood its purpose and how to provide accurate responses.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Universidade Nacional Timor Lorosa'e (UNTL), Dili Institute of Technology (DIT), Ministry of higher education and Culture Timor leste, Haburas foundation, Insitute Philosophy of Don bosco, Institue Nacional da Ciencia e Tecnologia(INCT), Komisaun Nasional (CN-UNESCO), Instituto Superior de Filosofia e de Teologia (ISFIT), Instituto de Ciências da Saúde (ICS), Instituto Nacional para o Desenvolvimento da Mão-de-Obra de Timor-Leste (INDMO IP), Hospital Nacional Guido Valadares (HNGV), Mahein foundation FNSP,Universidade da Paz (UNPAZ), Instituto São João de Brito (ISJB), TIC Timor, Instituto Profissional de Canossa (IPDC), Secretario Estado arte no Kultura(SEAC), Agência Noticiosa de Timor-Leste-TATOLI IP, MENZIES Timor-Leste, Instituto Nacional de Saúde Pública de Timor-Leste (INSPTL), Instituto da Defesa Nacional (IDN), Ministério dos Negócios Estrangeiros e da Cooperação (MNEC), MAPPF, Universidade Dili (UNDIL), Universidade Oriental de Timor Lorosa'e (UNITAL), Timor Furak, NATCOM.

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://untl.edu.tl/pt/>, <https://dit.edu.tl/>,
<https://mescc.gov.tl/en/>,<https://haburasfoundation.org/>,<https://www.tic.gov.tl/en/>,<https://www.idn.gov.pt/pt/>,https://www.menzies.edu.au/page/Research/Global_and_Tropical_Health/Timor-Leste/,<https://undil.tl/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- ☒ Public
- ☒ Private
- ☒ Civil Society
- ☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☐ Yes

☒ No

17. 1.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Open Science has not been fully implemented in Timor-Leste yet. The main obstacles include limited access to necessary resources and infrastructure, a lack of awareness and training among researchers and institutions, and challenges in adapting traditional research practices to open access models. There are also institutional barriers, such as the slow adoption of open policies and resistance to change. However, plans are in place to address these issues. Stakeholders, including the Ministry of Higher Education, INCT, and other relevant organizations, are working on developing training programs, improving infrastructure, and creating policies to support the adoption of Open Science in the country.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☐ Yes

☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☐ Yes

☒ No

20. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

21. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Yes, specific actions have been undertaken or are planned to be undertaken by the end of 2025 in Timor-Leste to incorporate the values and principles of open science in publicly funded research. The National Institute of Science and Technology (INCT) plays a key role in these efforts. INCT distributes funding for research and is creating a digital repository to disseminate research results to the public. This initiative is designed to promote open access, allowing research data, publications, and findings from publicly funded projects to be widely accessible, supporting transparency and public engagement in the research process. Additionally, the National University of Timor-Leste (UNTL) is developing institutional repositories and platforms for academic research, further promoting open access within the academic community. The Department of Information Technology (DIT) also plays a role by working on infrastructure development and systems that can support the sharing of research data and outputs across different institutions and stakeholders. Other institutions in Timor-Leste are also contributing to the development of a more open science ecosystem, including universities and research organizations that are building their own repositories and digital platforms to host and share research results. These efforts collectively aim to create a more accessible and collaborative research environment, facilitating the integration of open science values in publicly funded research by 2025.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

22. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☐ Yes

☒ No

23. 2.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, Timor-Leste does not yet have a national policy, strategy, or plan of action on science, technology, and innovation (STI). The absence of such a framework is due to various challenges, including limited financial resources for research and development (R&D), insufficient institutional capacity, and a lack of coordination among key stakeholders. Additionally, STI development has not been a primary national priority, as the country has focused on addressing fundamental socio-economic issues such as infrastructure, healthcare, and education. However, steps are being taken to address this gap. The National Institute of Science and Technology (INCT), in collaboration with the Organisation of African, Caribbean, and Pacific States (OACPS) through the Policy Support Facility (PSF), is currently working on the elaboration of an STI policy for Timor-Leste in 2024-2025. This initiative aims to provide a strategic framework to strengthen the country's research ecosystem, support innovation, and enhance international collaboration in STI. While this process is still in its early stages, it marks an important step toward the formalization of STI governance in Timor-Leste. Continued investment, capacity-building, and multi-stakeholder engagement will be essential to ensure the successful development and implementation of this policy.

24. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☐ Partly
- ☒ No

25. 2.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

No, Timor-Leste does not currently have a dedicated policy, set of policies, or legal framework at the national level that specifically addresses open science or its key elements in line with the 2021 UNESCO Recommendation on Open Science. The absence of such a framework is primarily due to the country's still-developing research ecosystem, limited institutional capacity, and lack of technical infrastructure to support open science practices. One of the main challenges is the limited digital and research infrastructure, which makes it difficult to establish open-access repositories, data-sharing platforms, and collaborative research networks. Additionally, funding constraints have hindered investment in open science initiatives, as research funding in Timor-Leste is largely dependent on external sources, often tied to specific donor-driven priorities rather than a national open science agenda. Another key obstacle is the lack of awareness and technical expertise on open science principles among researchers and policymakers. Since the country's scientific community is still relatively small, there has been limited advocacy for the development of open science policies. Furthermore, the absence of a national STI policy has contributed to the delay in formalizing open science governance. Despite these challenges, there are emerging efforts that could contribute to future open science policy development. The National Institute of Science and Technology (INCT) has been actively working on improving research dissemination, including the creation of research repositories. Additionally, INCT's collaboration with the OACPS-PSF to develop an STI policy in 2024-2025 may provide an opportunity to incorporate open science principles into the broader national research framework. However, further investment, capacity-building, and international partnerships will be necessary to establish a comprehensive national policy on open science in the future.

26. 2.3 In your country, are there specific policy instruments^[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☐ Yes
- ☒ Under development
- ☐ No

27. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

The National Institute of Science and Technology (INCT) of Timor-Leste is leading key initiatives to strengthen the country's Science, Technology, and Innovation (STI) policy and establish a national digital repository. These efforts aim to build a robust research and innovation ecosystem, enhance access to scientific knowledge, and drive technological advancement. INCT plays a central role in shaping and implementing Timor-Leste's STI policy, focusing on:

- Fostering Innovation across sectors like agriculture, education, health, and the environment.
- Promoting Research and Development (R&D) to enhance scientific capacity and collaboration.
- Building Institutional Capacity to strengthen national research institutions and universities.
- Encouraging International Collaboration to connect Timor-Leste with global research networks.
- Advancing Open Science Principles through open access to research and data.

Next, a key component of the STI policy, the national digital repository will serve as a centralized platform for storing, managing, and sharing research publications, data, and educational resources. Key features include:

- Open Access to Research to make Timorese scientific outputs widely available.
- Data Sharing and Management for greater transparency and reuse.
- Collaboration and Capacity Building to support researchers and institutions in digital research practices.
- Integration with International Platforms to align with global open science standards.

Challenges and Opportunities : While infrastructure, capacity building, and sustainability pose challenges, opportunities exist in global collaboration, local innovation promotion, and inclusive development. The success of these initiatives will depend on strong partnerships, sustainable funding strategies, and ongoing policy support. By advancing STI policies and open science initiatives, INCT is positioning Timor-Leste to strengthen its research landscape, foster innovation, and contribute to national and global scientific progress.

28. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☒ Under development
- ☐ No

29. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

In Timor-Leste, a national implementation plan, strategy, or policy for open science is currently under development as part of broader efforts to strengthen the country's research and innovation ecosystem. This initiative is being led by the National Institute of Science and Technology (INCT) in collaboration with the Organisation of African, Caribbean, and Pacific States (OACPS) Policy Support Facility (PSF) through the ongoing Science, Technology, and Innovation (STI) policy development process for 2024-2025. The STI policy aims to establish a foundation for research governance, which includes provisions for open science principles such as open access to research outputs, knowledge-sharing, and capacity-building for researchers. While no specific open science roadmap has been finalized yet, this process represents a significant step toward integrating open science into national policies. Key stakeholders involved in this initiative include the Ministry of Higher Education, Science, and Culture (MESCC), which oversees research and higher education institutions, and various universities and research institutions, such as the National University of Timor-Leste (UNTL) and Dili Institute of Technology (DIT), which are working on institutional repositories for research dissemination. While the development of open science policy is still in its early stages, Timor-Leste's participation in international discussions and collaborations through OACPS and UNESCO frameworks provides an opportunity to align national efforts with global best practices. However, continued investment, infrastructure development, and policy coordination will be necessary to ensure effective implementation in the coming years.

30. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (*ref.: (ii) 17.d, 17.e, 17.g*)

- ☐ Yes
- ☒ Under development
- ☐ No

31. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☐ Yes
- ☒ Under development
- ☐ No

32. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

In Timor-Leste, specific funding mechanisms for open science are currently under development, though they remain in the early stages. There is no dedicated national funding program explicitly supporting open science initiatives, but efforts are being made to integrate open science principles into broader research funding strategies. The National Institute of Science and Technology (INCT), as the primary research agency, has been selecting and funding research proposals from national researchers, with a focus on research dissemination and accessibility. While not explicitly framed as open science funding, these efforts contribute to increasing public access to scientific knowledge. Additionally, the ongoing development of the Science, Technology, and Innovation (STI) policy (2024-2025), in collaboration with OACPS-PSF, presents an opportunity to establish structured funding mechanisms that support open-access research, data-sharing, and collaboration. At the institutional level, universities such as UNTL and DIT have been working on repository systems for research outputs, but these initiatives largely rely on internal budgets and external partnerships rather than a dedicated national funding stream for open science. The main challenges in developing open science funding mechanisms include limited government resources for research, lack of dedicated budget allocations, and reliance on external funding for scientific activities. However, as Timor-Leste strengthens its research governance, the establishment of funding mechanisms specifically designed for open science is expected to be considered within broader national research development plans.

33. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☐ Yes

☒ No

34. 2.7 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Timor-Leste, open science practices have not yet been integrated into existing research funding mechanisms. The National Institute of Science and Technology (INCT), which is responsible for selecting and funding research proposals, receives funding directly from the government to support research activities. However, there are currently no specific funding conditions that mandate or incentivize open science practices such as open-access publishing, data-sharing, or collaborative research beyond traditional dissemination efforts. One of the main challenges is the lack of a formal policy framework requiring open science integration within research funding. Additionally, limited awareness and technical infrastructure for managing open-access research outputs contribute to the slow adoption of open science practices in national funding mechanisms. Despite these challenges, Timor-Leste is developing a Science, Technology, and Innovation (STI) policy (2024-2025), which may include provisions for incorporating open science into national research funding. As research governance structures evolve, the integration of open science principles into funding criteria is expected to be considered in future policy discussions.

35. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

36. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In Timor-Leste, there have been no efforts to foster equitable public-private partnerships on open science in line with the 2021 UNESCO Recommendation on Open Science. Currently, collaboration between public research institutions and the private sector remains limited, and open science is not yet a structured component of national research and innovation initiatives. Several challenges hinder the development of such partnerships, including the absence of a formal open science policy, a lack of awareness among private sector stakeholders, and limited research investment from private enterprises. Additionally, most scientific research in Timor-Leste is government-funded through institutions like the National Institute of Science and Technology (INCT), with minimal private sector engagement in research activities. Despite these obstacles, the ongoing development of a Science, Technology, and Innovation (STI) policy (2024-2025) could create opportunities to establish mechanisms that encourage public-private collaborations in research, including potential incentives for private sector involvement in open science initiatives. However, no concrete plans have been outlined for fostering such partnerships by 2025.

37. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☐ Yes

☒ Under development

☐ No

38. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

In Timor-Leste, a national monitoring framework for open science is currently under development. As part of the broader effort to establish a Science, Technology, and Innovation (STI) policy for 2024-2025, discussions have included the need to integrate monitoring mechanisms and specific indicators for open science within the national research and innovation landscape. The National Institute of Science and Technology (INCT), in collaboration with OACPS-PSF, is leading the development of the STI policy, which is expected to include open science-related metrics to track research output, accessibility, and collaboration. However, no formal indicators or evaluation frameworks have been finalized at this stage. Key challenges in implementing an open science monitoring system include limited technical capacity, lack of standardized data collection methods, and insufficient digital infrastructure to track open-access research outputs. Future developments in national STI governance may address these gaps, but as of now, open science monitoring remains in the early stages of formulation.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

39. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

There is no official data available on Timor-Leste's research and development (R&D) expenditure as a percentage of its Gross Domestic Product (GDP).

40. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

As of January 2024, approximately 742,400 individuals, representing 54.2% of Timor-Leste's population, had access to the internet (<https://datareportal.com/reports/digital-2024-timor-leste>). This data was published in February 2024, reflecting internet usage at the start of that year. Regarding bandwidth availability, Timor-Leste has historically faced challenges with slow internet speeds. In January 2024, research indicated that the country had the fourth-slowest internet speeds globally, just ahead of Afghanistan, Yemen, and Syria. However, significant infrastructure developments are underway to improve connectivity. The government has approved the construction of a submarine cable link to Australia, aiming to enhance international bandwidth and overall internet speed. Additionally, Telemor, one of the country's major telecommunications providers, plans to launch a 5G service once the submarine cable is operational, which is expected to meet the increasing data demand and improve bandwidth availability. These initiatives are part of Timor-Leste's broader strategy to expand digital services and improve internet infrastructure, as outlined in the Timor Digital 2032 plan.

41. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☐ Yes

☒ No

42. 3.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

Timor-Leste does not yet have a well-established National Research and Education Network (NREN) due to several key challenges. The country's digital infrastructure, including internet connectivity and broadband access, is still in the developmental stage, with slow internet speeds and limited bandwidth hindering the creation of a robust NREN. Financial constraints also play a significant role, as Timor-Leste has limited resources to invest in large-scale infrastructure projects like an NREN. Additionally, there is a shortage of specialized technical expertise needed to design, implement, and maintain such networks. Geographically, Timor-Leste faces challenges due to its scattered population and remote areas, making widespread, reliable internet access difficult. Furthermore, there is no coordinated national strategy or policy in place specifically aimed at developing an NREN. However, the government's Timor Digital 2032 plan, which includes improving broadband infrastructure, establishing a submarine cable link to Australia, and rolling out 5G services, could eventually provide the necessary foundation for an NREN in the future.

43. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☐ Yes
- ☒ Under development
- ☐ No

44. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☐ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- ☐ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

45. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

In Timor-Leste, national and institutional open science infrastructures are still under development. One of the key initiatives is the creation of a national digital repository by the National Institute of Science and Technology (INCT). This effort is focused on building a centralized platform to host and share scientific research and data. Alongside INCT, several academic and research institutions are also contributing to the development of open science infrastructures through their own digital repositories and platforms: National University of Timor-Leste (UNTL) - UNTL has developed institutional repositories for hosting academic publications and research materials. University of Díli (UNDIL) - UNDIL is working on making its academic research accessible and may eventually contribute to open science initiatives. Department of Information Technology (DIT) - The DIT is developing digital platforms and systems that could eventually support open science infrastructure. Timor-Leste National Library - While its focus is on providing general access to information, it plays a role in archiving and digitizing materials that could contribute to open access in the country. These institutions, along with the government's digital transformation efforts, are laying the foundation for a more comprehensive open science ecosystem in Timor-Leste, though much of the development is still in progress.

46. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☒ No

47. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

☐ Yes

☒ No

☐ Other

48. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

49. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☐ Yes

☒ No

50. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, systematic capacity building on open science has not yet taken place in higher education and research-performing institutions in Timor-Leste. Several factors contribute to this, including limited awareness and understanding of open science practices within many academic and research institutions. Open science, which involves practices like open access, open data, and open collaboration, is still a relatively new concept in the country. Additionally, there is a shortage of specialized expertise and training resources in this area. Timor-Leste also faces challenges with its digital infrastructure, including slow internet speeds, limited access to high-bandwidth networks, and insufficient research platforms, which make it difficult for institutions to fully engage in open science practices. Financial constraints further limit the country's ability to invest in programs that would build capacity for open science, such as training, workshops, and the development of necessary infrastructure. Moreover, there is limited institutional support for systematic capacity building in open science, and efforts in this area are often fragmented and ad-hoc. However, with ongoing initiatives such as the Timor Digital 2032 plan to improve digital infrastructure, there are future opportunities for capacity building in open science. As infrastructure and resources improve, there may be a stronger foundation for implementing systematic open science training programs in the country.

51. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

52. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, capacity building programs for policymakers on how to integrate the core values and principles of open science into Science, Technology, and Innovation (STI) policies and strategies have not yet taken place in Timor-Leste. Several reasons contribute to this; firstly, there is limited awareness and understanding of open science among policymakers. Open science practices, which include open access, open data, and collaborative research, are still emerging concepts in the country, and the integration of these principles into national policies has not been a priority. Secondly, Timor-Leste faces significant challenges with its digital infrastructure, which has limited the government's ability to fully engage with or promote open science. Without a solid digital foundation, policymakers may not fully recognize the value or feasibility of incorporating open science principles into STI strategies. Additionally, there are financial constraints, which hinder the development of specific capacity-building programs targeting policymakers. Limited resources also impact the creation of the necessary tools, platforms, and training programs to support such initiatives. While no formal capacity-building programs have taken place, Timor-Leste's ongoing efforts to improve digital infrastructure and enhance its research and education environment could lay the groundwork for future initiatives. As digital and policy landscapes improve, there may be opportunities to develop capacity-building programs for policymakers to better understand and integrate open science into STI policies and strategies.

53. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

54. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, a framework of open science competencies has not been incorporated into higher education research skills curricula in Timor-Leste, nor is it foreseen by the end of 2025. Several factors contribute to this situation; firstly, there is a ; of awareness and understanding of open science principles among educators and institutions in Timor-Leste. Open science practices, such as open access, open data, and collaborative research, are still emerging concepts in the country, and they have not yet been integrated into the academic curriculum. Secondly, Timor-Leste faces challenges with its digital infrastructure, which limits access to the tools and resources necessary to teach open science practices effectively. Slow internet speeds and limited access to high-bandwidth networks further complicate the development of digital literacy in research and open science. Moreover, there are financial and resource constraints, which hinder the development and implementation of comprehensive curricula on open science. This includes limited funding for training programs and the lack of specialized teaching materials and platforms for open science education. Although no formal framework for open science competencies currently exists in higher education curricula, the government's digital transformation efforts, such as Timor-Leste Digital 2032 plan, could create opportunities for future integration. As infrastructure and awareness improve, there may be plans to introduce open science principles into research skills curricula by the end of 2025 or shortly thereafter

55. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☐ Partly
- ☒ No

56. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)



Yes



No

57. 4.5 cont. **If yes**, please provide more information and links as relevant.

Yes, Timor-Leste has been making efforts to promote open science through the development of digital repositories that enhance access to scientific knowledge. The Instituto Nacional de Ciências e Tecnologia (INCT) supported by the Organisation of African, Caribbean, and Pacific States (OACPS) is leading an initiative to establish a national digital repository aimed at centralizing research outputs from various institutions. It seeks to improve collaboration among researchers, ensure the long-term preservation of scientific data, and facilitate open access to knowledge. Alongside INCT's efforts, several higher education institutions have also been developing their own research repositories. The National University of Timor-Leste (UNTL) has been working on an institutional repository to store and provide access to academic research, theses, and publications produced by students and faculty members. Similarly, the Dili Institute of Technology (DIT) has developed a repository system focused on archiving research projects and technical studies in engineering, technology, and development. Other universities and research centers in Timor-Leste are also adopting digital archiving systems to support knowledge dissemination and increase research visibility. These initiatives reflect Timor-Leste's commitment to strengthening its research infrastructure and fostering open science. By making scientific knowledge more accessible, these repositories contribute to interdisciplinary collaboration, informed policymaking, and the overall advancement of science and technology in the country.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

58. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☐ Yes

☒ No

59. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There have been no national or institutional initiatives in Timor-Leste to review existing research assessment and evaluation processes in alignment with open science principles and values. Several challenges hinder progress in this area. First, the country's research ecosystem is still in its early stages of development, with limited institutional frameworks for research evaluation. The National Institute of Science and Technology (INCT) is in the process of establishing research policies, but there is no structured mechanism for assessing research performance based on open science criteria. Second, universities and research institutions, such as the National University of Timor-Leste (UNTL) and Dili Institute of Technology (DIT), face resource constraints, including a lack of funding, digital infrastructure, and trained personnel to implement new research evaluation systems. Many researchers still rely on traditional assessment methods, often based on academic theses and project reports, rather than open-access publications, data sharing, or public engagement. Additionally, awareness and policy integration of open science remain limited. While there is growing interest in open access initiatives, aligning research evaluation with open science requires policy reforms, institutional coordination, and technical capacity-building, which have yet to be prioritized. At present, there are no concrete plans to review research assessment processes by the end of 2025. However, as Timor-Leste continues to develop its science and innovation policies, future efforts may focus on integrating open science principles into research evaluation frameworks.

60. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

61. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Currently, there are no specific clauses recognizing open science practices, such as sharing, collaboration, engagement beyond the scientific community, or dialogue with other knowledge systems, in career evaluation and progression systems in Timor-Leste. Several challenges contribute to this situation. Firstly, research career evaluation frameworks are still underdeveloped in the country. Academic institutions, such as the National University of Timor-Leste (UNTL) and Dili Institute of Technology (DIT), primarily assess faculty and researchers based on traditional criteria, such as teaching performance and completed research projects, rather than open science contributions. There is no standardized mechanism to evaluate researchers based on open-access publishing, data sharing, or engagement with non-academic stakeholders. Secondly, the absence of a national research funding system and limited institutional support for scientific research create obstacles to recognizing and rewarding open science practices. Researchers often lack incentives to engage in interdisciplinary collaboration, open-access dissemination, or public engagement, as these efforts are not yet formally valued in career progression policies. The National Institute of Science and Technology (INCT), which plays a central role in selecting research proposals, supporting research activities, and disseminating research outputs, has not yet incorporated open science criteria into its research evaluation processes. While INCT promotes knowledge sharing through dissemination efforts, there are no specific policies linking open science practices to researchers' career advancement. Timor-Leste is in the process of developing its Science, Technology, and Innovation (STI) policy, which may provide opportunities to integrate open science principles into national research governance. However, at present, there are no formal plans to incorporate open science practices into career evaluation and progression systems by the end of 2025. Future policy developments, particularly under the STI framework, may consider addressing this gap as the country strengthens its research ecosystem.

62. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☒ Yes

☐ No

63. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

Yes, there have been some initiatives in Timor-Leste to recognize and promote open science practices, although they are still in their early stages. One key initiative is the establishment of digital research repositories by the National Institute of Science and Technology (INCT) and academic institutions such as the National University of Timor-Leste (UNTL) and Dili Institute of Technology (DIT). These repositories aim to facilitate open access to research outputs, enabling knowledge sharing among researchers, policymakers, and the public. INCT, as the main research agency, has been supporting researchers by selecting research proposals and disseminating research findings, contributing to broader access to scientific knowledge. At the university level, UNTL and DIT have taken steps to develop institutional repositories where students and researchers can store and share their research work. These efforts align with open science principles by promoting accessibility and collaboration. Although these initiatives do not yet include formal reward mechanisms, they represent a shift toward recognizing the value of open science. The ongoing development of the Science, Technology, and Innovation (STI) Policy in Timor-Leste may further institutionalize open science practices, potentially leading to structured incentives or recognition systems for researchers engaging in open access, data sharing, and interdisciplinary collaboration. For now, these initiatives remain limited in scope, and further investment in digital infrastructure, policy frameworks, and capacity-building is necessary to fully integrate open science into the national research system.

64. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.



Yes



No

65. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

Yes, there have been some unintended negative consequences of open science practices reported in Timor-Leste, primarily related to the lack of a robust legal and institutional framework to regulate research dissemination and data sharing. One of the main concerns is the risk of misuse or misinterpretation of research findings. Since open access allows unrestricted access to research outputs, there have been instances where data and findings were used without proper context, leading to potential misrepresentation of scientific knowledge. This is particularly relevant in areas such as public health and environmental research, where inaccurate interpretations could influence policy decisions or public perception. Another challenge is the lack of intellectual property (IP) protection and recognition for researchers. Many researchers in Timor-Leste have expressed concerns about their work being accessed and used without proper citation or acknowledgment. The absence of clear policies on research ownership and data governance increases the risk of unauthorized use of scientific outputs. Furthermore, limited digital infrastructure and cybersecurity risks pose challenges to open science practices. As institutions such as INCT, UNTL, and DIT develop digital repositories, concerns about data security, privacy, and the long-term sustainability of these platforms have been raised. Without proper safeguards, there is a risk of data loss or unauthorized access to sensitive research information. Currently, Timor-Leste does not have a formal strategy or plan to prevent and mitigate these consequences. However, the ongoing development of the Science, Technology, and Innovation (STI) Policy presents an opportunity to address these issues by establishing guidelines for open science, research ethics, data protection, and intellectual property rights. Strengthening institutional frameworks and providing training for researchers on responsible open science practices will also be key steps in mitigating these risks.

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

66. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (*ref.: (vi) 21*)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☒ No

67. 6.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, there have not been any significant national or institutional initiatives in Timor-Leste that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process. Several challenges contribute to this situation. Firstly, scientific research infrastructure and funding remain limited, making it difficult to implement systematic changes that promote open science across all research stages. Most research activities in the country are still conducted on a small scale, often relying on external funding, which influences the methodologies and dissemination strategies used. Secondly, there is no established national framework for open science governance. While institutions such as the National Institute of Science and Technology (INCT), the National University of Timor-Leste (UNTL), and Dili Institute of Technology (DIT) support research activities, they do not yet have specific policies that encourage participatory or transparent research processes, such as open peer review, citizen science, or collaborative research platforms. Additionally, digital infrastructure and technical capacity remain underdeveloped, limiting the ability of researchers to engage in open data-sharing practices, interactive research methodologies, and collaborative knowledge production. Without adequate training and resources, the adoption of open science methodologies remains slow. At present, there are no formal plans to introduce such initiatives by the end of 2025. However, as the Science, Technology, and Innovation (STI) Policy continues to develop, there may be opportunities to integrate participatory and transparent approaches into the national research system. Capacity-building efforts, institutional reforms, and digital advancements will be necessary to facilitate the transition toward a more open and collaborative research environment in Timor-Leste.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL
AND KNOWLEDGE GAPS

68. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Timor-Leste is in the early stages of promoting and facilitating international scientific collaborations on open science, in alignment with the 2021 UNESCO Recommendation on Open Science. While there is no dedicated national policy on open science, several initiatives contribute to fostering international cooperation in this area.

1. Engagement through the National Institute of Science and Technology (INCT) The National Institute of Science and Technology (INCT) plays a central role in supporting research activities and facilitating collaborations with international partners. While INCT does not yet have a formal open science policy, it promotes knowledge dissemination by selecting and supporting research projects, including those involving international cooperation. INCT has also been involved in discussions about developing a Science, Technology, and Innovation (STI) Policy, which may incorporate open science principles in the future.

2. Institutional Partnerships and Open Research Repositories Academic institutions such as the National University of Timor-Leste (UNTL) and Dili Institute of Technology (DIT) have engaged in research collaborations with foreign universities and international research organizations. These collaborations often involve knowledge exchange, joint research projects, and the use of open-access platforms to share research findings. Institutional digital repositories are being developed to make research outputs more accessible, aligning with open science principles.

3. Participation in Regional and International Research Networks Timor-Leste has engaged with regional and international research networks, particularly in areas such as public health, environmental sustainability, and climate change. These collaborations often involve data-sharing agreements and joint publications, although there is still a need to formalize open science policies at the national level. The country's engagement with ASEAN, CPLP (Community of Portuguese-Speaking Countries), and UN research initiatives provides opportunities to expand open science practices.

Challenges and Future Directions : Despite these efforts, Timor-Leste faces several challenges in fully implementing open science collaborations, including: (a) limited digital infrastructure and research funding to support open-access platforms and data-sharing initiatives; (b) lack of national policies and regulations that formally integrate open science principles into research governance; (c) capacity-building needs, as many researchers and institutions require training on open science methodologies and tools. To address these challenges, Timor-Leste is working on the STI Policy, which may provide a framework for promoting open science and expanding international collaborations. Future efforts could focus on establishing open science agreements with international research institutions, increasing investment in research infrastructure, and developing national guidelines for open-access publishing and data-sharing.

69. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

70. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, Timor-Leste does not currently have a strategy or plan for stimulating cross-border, multi-stakeholder collaboration on open science, including best practice exchange, capacity building, and open science metrics. The absence of such a strategy is primarily due to the lack of a national open science policy or regulatory framework that establishes guidelines for international collaboration. Additionally, research infrastructure and funding remain limited, making it difficult to support open-access platforms, data-sharing initiatives, and capacity-building programs. Many researchers and institutions in the country also have limited awareness of open science practices, and there are few structured training initiatives to address this gap. Furthermore, collaboration between stakeholders, including universities such as UNTL and DIT and institutions like INCT, remains informal and uncoordinated, with no centralized mechanism for engagement. While there are no immediate plans to introduce a structured strategy, the ongoing development of the Science, Technology, and Innovation (STI) Policy may provide a foundation for future initiatives. This could include fostering international research collaborations, developing training programs on open science, and establishing national open-access repositories to improve data-sharing and transparency. However, significant investment in research infrastructure and institutional capacity will be necessary for Timor-Leste to advance in this area.

71. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

72. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No, Timor-Leste is not currently involved in discussions on the establishment of regional and international funding mechanisms for open science. This is largely due to the country's limited participation in global and regional research funding frameworks, as well as the absence of a national open science policy that could facilitate engagement in such discussions. Additionally, the research ecosystem in Timor-Leste is still developing, with most funding for scientific research coming from external sources, often allocated through bilateral or multilateral agreements rather than dedicated open science funding mechanisms. The lack of institutional capacity and technical expertise in managing open-access research funding further hinders participation in these discussions. While there are no immediate plans to engage in regional or international funding initiatives specifically for open science, ongoing efforts to develop the Science, Technology, and Innovation (STI) Policy may create opportunities for future engagement. Strengthening research governance, increasing investment in national research programs, and fostering international partnerships will be essential steps toward positioning Timor-Leste to participate in open science funding mechanisms in the future.

Part 8. GENERAL CONSIDERATIONS

73. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

74. 8.1 cont. If yes, please provide more information.

Yes, several external factors may impede the implementation of the 2021 UNESCO Recommendation on Open Science in Timor-Leste. One of the primary challenges is the limited international research collaboration and funding opportunities specific to open science. As a small developing nation, Timor-Leste relies heavily on external funding for research, which is often tied to specific donor priorities rather than open science principles. Without dedicated financial mechanisms, the country struggles to invest in the necessary infrastructure for open access and data-sharing initiatives. Another major obstacle is the lack of regional frameworks for open science collaboration in Southeast Asia that explicitly include Timor-Leste. While some regional organizations promote scientific cooperation, Timor-Leste is not yet fully integrated into ASEAN's science and technology initiatives, which limits opportunities for participation in broader open science networks and capacity-building programs. Furthermore, digital infrastructure and internet connectivity challenges pose significant barriers to the development of open-access repositories and data-sharing platforms. Many research institutions in Timor-Leste do not have the technical resources to implement large-scale open science initiatives, making it difficult to align with global standards. Lastly, language barriers and access to global scientific literature create additional difficulties. Much of the research published under open science frameworks is in English, whereas Timor-Leste primarily uses Tetum and Portuguese, limiting accessibility and engagement with international open science materials. While these external challenges hinder implementation, Timor-Leste's ongoing efforts to strengthen its Science, Technology, and Innovation (STI) Policy and develop research institutions such as the National Institute of Science and Technology (INCT) may create pathways for future alignment with UNESCO's open science recommendations. However, stronger regional partnerships, increased funding, and improved infrastructure will be necessary to overcome these external limitations.

75. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org